Ryan Schmelzle

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Summary

Computer vision and graphics researcher specializing in neural scene reconstruction, 3D perception, and rendering systems for telepresence. Experience developing robotic teleoperation systems and neural rendering pipelines for scene representation.
Interested in roles in computer vision, graphics, robotics, and neural rendering.

Experience

Graduate Researcher Jan. 2025 – Present
IRON Labs, UNC Chapel Hill

- Developing a robotic teleoperation system for neural scene reconstruction in search-and-rescue applications, deployed on a Boston Dynamics SPOT quadruped.
- Designing and evaluating learning-based scene representations (NeRFs, gaussian splats) to support visual inspection, remote navigation, and situational awareness in time-sensitive tasks or inhospitable environments.
- Built vision-analysis systems for reconstruction accuracy and object salience in NeRF and Gaussian Splat representations.
- Investigating alignment between robot perception outputs, human intent, and operation goals to ensure reliable teleoperation in time-critical situations.

Graduate Researcher Sept. 2023 – Present
Visual Computing & Augmented Intelligence Lab, UNC Chapel Hill

- Led development of prototype AR glasses integrating computer vision and real-time digital signal processing techniques for enhanced perception and audio focusing. Ongoing work in hardware deployment.
- Designed and implemented a spatial audio filtering method for high-speed beamforming and targeted listening.

Undergraduate Researcher Aug. 2022 – May. 2023
Graphics and Virtual Reality Group, UNC Chapel Hill

- Contributed to neural rendering research for high-speed telepresence on sparse RGB-D camera setups under Henry Fuchs.
- Assisted with dataset preparation, baseline implementation, and experimental evaluation; co-authored an IEEE TVCG paper on neural telepresence systems.

Software Engineering Intern May. 2022 – Aug. 2022
Ford Motor Company (Remote), Dearborn MI

- Developed components for a data-driven vehicle recommendation platform for large-scale user interaction.
- Implemented backend services, frontend interfaces and database solutions using Java, JavaScript, SQL, and Vue.js.
- Worked within CI/CD pipelines and code-quality frameworks (Git, Jenkins, SonarQube).

Projects

Parallax Display & GPU Ray Tracer 2024 – 2025
Personal Project

- Built a GPU ray tracer with BVH acceleration, Phong shading, shadow rays, specular reflections, Monte Carlo global illumination, antialiasing (FXAA, SSAA) and an interactive viewer.
- Designed real-time display system that turns a monitor into a 3D window using head tracking, off-axis perspective projection, and dynamic shadows to simulate occlusions and parallax shift from the viewer's perspective.
- Technologies: Numba CUDA, PyQt5, OpenCV (YuNet face detection, MiDaS depth estimation).

Monocular Depth Estimation *Personal Project* - 2024

- Designed a U-Net encoder-decoder with 3-stage iterative refinement to predict depth maps from RGB images.
- Custom pipeline with depth backbone (MSE, gradient matching, SSIM) and geometric constraints consistency(superpixel consistency, surface normal fidelity). Final per-pixel refinment using skip connections and latent RGB features.
- Trained on NYU Depth V2.

Education

University of North Carolina at Chapel Hill Chapel Hill, NC

Ph.D. Candidate, Computer Science Aug. 2023 - Expected May 2028
Focus: Neural rendering, computer vision, telerobotics

M.S. in Computer Science Aug. 2023 - Dec. 2024
Focus: Computational perception, audio-visual processing systems

B.S. in Computer Science & Communication Studies Aug. 2019 - May 2023
GPA: 3.92 / 4.0 Phi Beta Kappa; Honors Carolina; Highest Distinction

Relevant Courses: Computer Vision, Graphics & VR, Neural Rendering, Deep Learning, Computational Geometry

Skills

Core Proficiencies: Computer vision, neural rendering, NeRFs, Gaussian Splatting, 3D reconstruction, deep learning
Tools & Languages: Python, C++, SQL; PyTorch, OpenCV; ROS/ROS2, SLAM; Git, Linux; Blender